

Chapter Overview

- Key topics include zoo geographic regions, biodiversity, endangered species, reasons for wildlife depletion, conservation remedies, and major protection projects

SECTION 1 Zoo Geographic Regions of India

- India is divided into 9 zoo geographic regions based on regional climate and vegetation characteristics
- Himalayan region with cold altitude habitats
- Ladakh and dry cold area
- Forested region in lower Himalayas
- Highlands without forest cover in upper Himalayas
- Northern fertile plains
- Desert region of Rajasthan
- Peninsular plateau region
- Sea coasts and coastal ecosystems
- Nilgiri Hills region

SECTION 2 Biodiversity of India

- Around 15 lakh species of living organisms are recorded across the world
- India accounts for about 81,000 species of fauna ranking 12th in global biodiversity
- Asian elephants inhabit peninsular rain forests while one horned rhinos live in swampy Assam areas
- Snow leopards inhabit higher Himalayan ranges while wild goats live in Kashmir region
- Dense forests of Western Ghats harbor rare species like flying squirrels
- Indian lion is exclusively found in the Gir forest of Gujarat
- Marine biodiversity includes rare species of corals, sea turtles, and marine mammals near Lakshadweep and Gulf of Kutch
- Rare birds like the Great Indian Bustard inhabit grasslands while Nicobar Dove is native to Nicobar Islands

SECTION 3 Wildlife of Gujarat

- Gir forest in Gujarat is the sole natural habitat of the Asiatic Lion in all of Asia
- The Wild Ass or Gudkhur is uniquely found in the Little Rann of Kutch
- Marine national park in the Gulf of Kutch hosts rare sea cucumbers, corals, octopuses, and dolphins
- Velavadar National Park provides protection to blackbucks and bustard birds
- Nalsarovar wetland sanctuary serves as a major winter habitat for migratory water birds
- Rare birds like Chiloptera can be spotted in the mountainous forests of Vijaynagar Taluka

SECTION 4 Species on the Verge of Extinction

- Indian cheetah is already extinct in the wild across India
- Tigers were once widespread across many states but are now restricted to specific protected reserves
- Asiatic lions were once found up to Iran but are now restricted solely to Gir forest
- Birds like the pink headed duck, mountain quail, and vulture species face severe extinction threats
- Freshwater dolphins of Ganges and Indus rivers face severe survival threats due to water pollution
- Sea turtles visiting coastal beaches for nesting face habitat destruction and marine pollution

SECTION 5 Reasons for Threat to Wildlife

- Human greed and rapid infrastructure development lead to clearing of natural habitats
- Uncontrolled deforestation deprives wild animals of food, shelter, and breeding grounds
- Poaching and illegal hunting for animal skins, meat, horns, teeth, and bones drastically reduce populations
- Heavy cattle grazing in forest areas destroys grass cover needed by wild herbivores
- Scarcity of natural prey drives wild carnivores into human settlements leading to human animal conflict
- Forest fires, climate change, and marine water pollution cause widespread loss of wildlife species

SECTION 6 Remedies for Wildlife Conservation

- Strict enforcement of forest protection laws with heavy legal penalties against illegal hunting
- Expanding protected forest cover and establishing interconnected ecological corridors
- Controlling overgrazing in forest reserve zones to preserve natural vegetation for herbivores
- Setting up dedicated fire management teams and observation towers to prevent forest fires
- Encouraging public participation through educational campaigns and school Eco Clubs
- Conducting regular wildlife censuses to monitor animal population trends and health

SECTION 7 Major Wildlife Protection Projects

- Project Tiger was launched in 1973 to protect tigers from poaching and habitat loss across dedicated reserves
- Project Lion was initiated in 1972 to conserve and increase Asiatic lion populations in Gir forest
- Project Elephant focuses on protecting wild elephants and securing their natural migration corridors
- Project Rhino aims to protect the single horned Indian rhinoceros in swampy regions like Kaziranga
- Crocodile Project protects saltwater and freshwater crocodile species from poaching
- Snow Leopard Project targets conservation of snow leopards in high altitude Himalayan regions
- Hangul Project focuses on preserving the rare Kashmir stag or Barasingha deer

SECTION 8 Protected Area Networks

- Sanctuaries allow limited human activities like livestock grazing with government permission
- National Parks offer strict protection where human entry, grazing, and resource extraction are completely prohibited
- Biosphere Reserves represent large ecosystems created under UNESCO guidelines to preserve overall regional biodiversity
- India has established over 500 wildlife sanctuaries, over 100 national parks, and 18 biosphere reserves